

Geodesic Path Systems on K_4 -Subdivisions

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Abstract

A consistent path system selects one path between every pair of vertices and is closed under taking subpaths. It is metrizable if all selected paths are simultaneously shortest for some positive edge weighting. We determine the metric realizability of every consistent path system on every subdivision of K_4 . Such a subdivision is metrizable if and only if it has no topological minor among $\Theta_{3,3,4}$, Z_Δ , $Z_\star$, Z_A , Z_B , where the four Z -graphs are explicit K_4 -subdivisions and Z_Δ is a new minimal obstruction. We also classify the neighborly variant, in which each edge is the selected path between its endpoints: the coordinate obstructions are precisely Z_Δ , $Z_\star$, Z_A , Z_B and $N_{233} = P_4(2, 3, 3)$.

The positive direction is constructive and largely theoretical. Consistency turns route choices into Ferrers relations; monotone potentials realize these relations, while a scalar calibration synchronizes cross-chain routes, same-chain detours, and terminal entrance cuts. A terminal-potential ear extension handles an infinite family containing the apparent core $P_4(2, 3, 2)$. The only computer-assisted inputs are two explicitly isolated finite propositions: 1,052 systems on two seven-vertex cores and 476 compatible terminal-order instances on two six-vertex cores. Each instance is accompanied by a static positive rational witness and an exhaustive, optimization-free exact verifier.

Keywords. graph metrizability; consistent path system; K_4 -subdivision; Ferrers relation; shortest paths; exact certificates.

MSC 2020. 05C12, 05C38, 05C85.

1 Introduction

Cizma and Linial introduced graph metrizability as a graph-theoretic form of geodesic geometry [4]. A path system $\mathcal{P}$ in a finite connected graph G selects a simple path P_{uv} for every unordered pair u, v . It is *consistent* if every subpath of a selected path is the selected path between its endpoints. A positive edge weighting w induces $\mathcal{P}$ if every P_{uv} is a shortest u - v path. The graph is *metrizable* when every consistent path system is induced by some positive weighting.

For a fixed system $\mathcal{P}$, realizability is a polyhedral feasibility question in edge-weight space. Writing $\chi(Q)$ for the incidence vector of a path Q , its metric cone is

$$\mathcal{M}(\mathcal{P}) = \{w \in \mathbb{R}_{>0}^{E(G)} : \langle \chi(P_{uv}) - \chi(Q), w \rangle \leq 0 \text{ for every } u, v \text{ and every } u\text{-}v \text{ path } Q\}.$$

Thus positive results amount to constructing a point of this cone from the combinatorics of $\mathcal{P}$, whereas a non-metrizability certificate is a positive linear dependence among path inequalities that contradicts $w(e) > 0$. This discrete-geometric viewpoint guides both halves of the classification.

Metrizability is closed under taking topological minors and therefore has a finite obstruction set $\mathcal{F}_M$ [4]. Determining that set is Open Problem 11.6 of that paper. The same paper asks which graphs have the weaker property that every *neighborly* consistent system—one satisfying $P_{uv} = uv$ for every edge uv —is metrizable (Open Problem 11.8). Subsequent work gives strong

global structure for metrizable graphs [2] and a complete classification of metrizable theta graphs [8]:

$$\Theta_{a,b,c} \text{ is metrizable} \iff \min\{a, b, c\} \leq 2 \text{ or } (a, b, c) = (3, 3, 3). \quad (1.1)$$

This paper resolves both questions on the full topological type of K_4 . For a vector $\ell = (\ell_{01}, \ell_{02}, \ell_{03}, \ell_{12}, \ell_{13}, \ell_{23}) \in \mathbb{Z}_{\geq 1}^6$, let $K_4(\ell)$ be obtained by replacing kernel edge ij with an internally disjoint path of length ℓ_{ij} . Put

$$\begin{aligned} Z_\Delta &= K_4(1, 1, 1, 2, 2, 3), & Z_\star &= K_4(1, 1, 2, 1, 2, 3), \\ Z_A &= K_4(1, 1, 2, 2, 2, 2), & Z_B &= K_4(1, 2, 2, 2, 2, 1). \end{aligned} \quad (1.2)$$

The long kernel edges of Z_Δ form a triangle, those of $Z_\star$ form a star, and the two short kernel edges of Z_A, Z_B are respectively adjacent and opposite. The last three graphs occur in the obstruction collection of [4]; Z_Δ is new.

Theorem 1.1 (Ordinary classification). *A subdivision G of K_4 is metrizable if and only if no member of*

$$\{\Theta_{3,3,4}, Z_\Delta, Z_\star, Z_A, Z_B\} \quad (1.3)$$

is a topological minor of G . Moreover, Z_Δ is topologically minimal non-metrizable.

Thus Theorem 1.1 determines exactly the intersection of the unknown set $\mathcal{F}_M$ with the K_4 topological type. It does not claim a complete determination of $\mathcal{F}_M$ or of the strict obstruction set $\mathcal{F}_{SM}$.

The neighborly class behaves differently: adding a chord can destroy a non-metrizable neighborly path system, so the topological-minor mechanism no longer applies. For three long kernel edges forming a path, write $P_4(p, q, r)$ when the consecutive lengths are p, q, r , with q on the middle kernel edge.

Theorem 1.2 (Neighborly classification). *A subdivision $G = K_4(\ell)$ has every neighborly consistent path system metrizable if and only if, after every relabelling of the four kernel vertices, ℓ fails to dominate the length vector of each graph in*

$$\{Z_\Delta, Z_\star, Z_A, Z_B, N_{233}\}, \quad N_{233} = P_4(2, 3, 3). \quad (1.4)$$

Equivalently, G contains no coordinate subdivision of a graph in (1.4). The graph N_{233} is coordinate-minimal within the K_4 topological type.

The distinction between (1.3) and (1.4) is genuine. For example, $K_4(1, 1, 2, 2, 1, 4)$ contains $\Theta_{3,3,4}$, but it satisfies the criterion of Theorem 1.2; hence every neighborly consistent system on it is metrizable.

The proof has three layers. First, obstruction avoidance reduces the six kernel lengths to a short list of support shapes. Second, the infinite positive families are represented by one- and two-dimensional Ferrers potentials. The hardest family, $P_4(p, 2, r)$, has five possible chordless routes between two suspended chains; consistency separates them into three global modes, while a single scalar-feasibility lemma synchronizes bridge positivity, entrance cuts, same-chain detours, and exclusion of the unused fifth route. Third, two seven-vertex ordinary cores and two six-vertex terminal-order cores are discharged by static exact certificates. The apparent eight-vertex ordinary core $P_4(2, 3, 2)$ is instead the symbolic long-ear case H_3 . The computer is therefore not used to infer or extrapolate any infinite pattern: the archived rational vectors establish only the two finite propositions explicitly isolated below.

1.1 Relation to prior work and contributions

Cizma and Linial introduced consistent path systems, proved topological-minor closure of metrizability, and exhibited the three previously known K_4 -type obstructions $Z_\star, Z_A, Z_B$ [4]. The global structural results of Chudnovsky, Cizma, and Linial explain why subdivisions of small kernels are a natural testing ground [2]. The only external classification used in the positive direction is the theta theorem (1.1) [8]. No conclusion about the full obstruction sets is imported from these works.

The new contributions of the present paper are separated in Table 1. This separation is useful because it identifies exactly where the two computer-assisted propositions enter: neither finite archive is used to guess or extrapolate an infinite family.

Table 1: Logical provenance of the ingredients in the two classifications.

ingredient	role	status in this paper
theta classification	handles blocks after deletion of an unused edge and detects $\Theta_{3,3,4}$	external input
$Z_\star, Z_A, Z_B$	known K_4 -type obstructions	external input
Z_Δ and N_{233}	new ordinary and neighborly certificates	explicit integer certificates
shape reductions	reduce six arbitrary kernel lengths to the positive families	proved combinatorially
Ferrers-potential theory	realizes the two-chain and split-bridge infinite families	symbolic and constructive
finite and terminal cores	closes two seven-vertex bases and one six-vertex terminal interface	static exact rational certificates

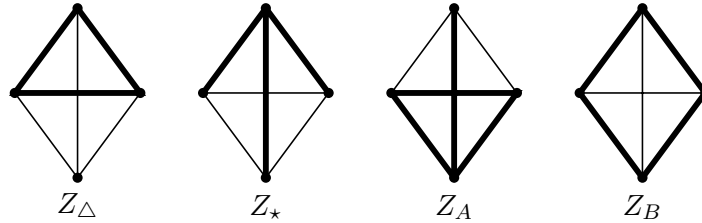


Figure 1: Long-edge supports of the four K_4 -type obstructions. A thick kernel edge has length at least two; in Z_Δ and $Z_\star$, one of the three thick edges has length three.

2 Preliminaries

All graphs are finite, simple, undirected, and connected. Paths are simple. For a path Q , its weight is denoted by $w(Q)$.

Definition 2.1. A path system $\mathcal{P} = \{P_{uv} : \{u, v\} \in \binom{V(G)}{2}\}$ is *consistent* if for any $x, y \in V(P_{uv})$, the x - y subpath of P_{uv} is P_{xy} . It is *neighborly* if $P_{uv} = uv$ for every edge uv . A positive weighting $w : E(G) \rightarrow \mathbb{R}_{>0}$ induces $\mathcal{P}$ if $w(P_{uv}) \leq w(Q)$ for every u - v path Q .

We only require non-strict induction. Several constructions below make all listed chordless comparisons strict, but no theorem in this paper asserts strict metrizability.

Lemma 2.2 (Unused-edge reduction). *Let $\mathcal{P}$ be consistent and let $e = xy \in E(G)$ with $P_{xy} \neq e$. Then no selected path uses e . If every block of $G - e$ is metrizable, then $\mathcal{P}$ is induced by a positive weighting of G .*

Proof. If e occurred in a selected path, consistency would make its one-edge subpath the selected x - y path, a contradiction. Restrict $\mathcal{P}$ to the blocks of $G - e$, choose inducing weights there, and multiply block weights successively so that their restrictions glue across cut vertices. This is the standard cut-vertex gluing argument: every path between different blocks is the concatenation of the selected paths to the separating cut vertices. Finally set $w(e) > \sum_{f \neq e} w(f)$. No shortest selected path uses e , and any path using e is longer than a path in $G - e$ with the same endpoints. $\square$

Lemma 2.3 (Chord elimination). *Let $\mathcal{P}$ be neighborly and let $w > 0$. If every selected path is no longer than every chordless competitor with the same endpoints, then w induces $\mathcal{P}$.*

Proof. Induct on the number of edges of a competing path Q . If Q has a chord xy , replace the x - y subpath by the edge $xy = P_{xy}$. By induction the replacement does not increase weight; after deleting a positive-weight closed walk if one is created, the number of edges decreases. Iteration reaches a chordless competitor. $\square$

Lemma 2.4 (Subdivision lift for neighborly obstructions). *If H has a non-metrizable neighborly consistent path system, then every subdivision of H has one.*

Proof. It suffices to subdivide uv once, into uzv . Expand every occurrence of uv in the old paths. For an old vertex x , define P_{zx} to be zuP_{ux} if P_{ux} does not pass through v , and zvP_{vx} otherwise. The resulting system is consistent and neighborly. Inducing weights would contract to an inducing weight of H by setting $w(uv) = w(uz) + w(zv)$, a contradiction. $\square$

The coordinate language in Theorem 1.2 is intrinsic to the K_4 topological type. We record the precise equivalence so that no containment assertion later depends on a chosen drawing.

Lemma 2.5 (Coordinate domination). *Let $G = K_4(\ell)$ and $H = K_4(h)$. Then H is a topological minor of G through a model that uses all four degree-three vertices if and only if there is a permutation $\pi \in S_4$ such that*

$$\ell_{\pi(i)\pi(j)} \geq h_{ij} \quad (0 \leq i < j \leq 3). \quad (2.1)$$

In particular, coordinate subdivision is independent of the initial names of the kernel vertices.

Proof. The degree-three vertices of any subdivision of K_4 are exactly its four kernel vertices. Hence a topological model using all four of them maps the kernel vertices of H bijectively to those of G , giving a permutation π . After that map is fixed, the six branch paths of the model are internally disjoint and their internal vertices have degree two. A branch path from $\pi(i)$ to $\pi(j)$ therefore lies on the unique subdivided kernel edge between those vertices. It can contain a subdivision of the corresponding path of H exactly when the first length is at least h_{ij} . This proves necessity. Conversely, under (2.1), choose $h_{ij} - 1$ internal vertices on the $\pi(i)$ - $\pi(j)$ corridor to represent the prescribed degree-two vertices of H ; any remaining vertices merely subdivide edges of H . The union of the six corridors is therefore a subdivision of H . $\square$

Lemma 2.6 (Proper-minor reduction inside the K_4 type). *Let $G = K_4(\ell)$, and let H be a proper topological minor of G . Then either*

- (a) *H is a coordinate-shortened K_4 -subdivision, after a kernel relabelling; or*
- (b) *every nontrivial block of H is contained in a graph obtained by breaking one kernel path of G . Such a block is a theta subdivision, a cycle, or a subgraph of one of these.*

Proof. Take a subdivision of H as a subgraph $H' \subseteq G$. If H' contains all four degree-three vertices and connects every pair by the six original kernel corridors, then suppression of degree-two vertices gives (a) by Lemma 2.5. Otherwise some kernel corridor is broken or a branch vertex is omitted. Delete a first missing edge of such a corridor. The two surviving corridor pieces are pendant outside the unique cyclic block, while the other five corridors form a theta graph between the endpoints of the deleted kernel edge. Omitting more edges or a branch vertex only takes subgraphs and splits blocks further. Thus every remaining nontrivial block has the form in (b). $\square$

We use three established facts. Metrizable is closed downward under topological minors, every outerplanar graph is metrizable [4], and the theta classification is (1.1) [8]. In particular, cycles are metrizable, and a theta graph avoiding $\Theta_{3,3,4}$ is metrizable.

3 Obstructions and exact certificates

3.1 The new ordinary obstruction Z_Δ

Let the kernel vertices be a, b, c, d . Replace the triangle abc by

$$a - p - b, \quad a - q - c, \quad b - r - s - c,$$

and retain ad, bd, cd . The following triangular table specifies all $\binom{8}{2} = 28$ unordered pairs. Destinations and paths in the last two columns are aligned entry by entry.

root	new destinations	selected paths
a	p, q, d, b, c, r, s	$ap, aq, ad, apb, aqc, apbr, aqcs$
b	p, r, s, d, q, c	$bp, br, brs, bd, bpaq, brsc$
c	q, s, r, d, p	$cq, cs, csr, cd, csrbp$
d	p, q, r, s	$dap, dcq, dbr, dbrs$
p	q, r, s	$paq, pbr, pbrs$
q	r, s	$qapbr, qcs$
r	s	rs

(3.1)

Each graph edge occurs as the selected path between its endpoints. Moreover, reading any subpath of any entry in (3.1) produces the entry for the corresponding pair. Thus the system is neighborly and consistent. The script `verify_ztriangle.py` independently reconstructs the table and checks these assertions before verifying the certificate below.

If w induces it, compare the following six selected paths with the displayed competitors:

$$\begin{aligned}
w(dbrs) &\leq w(dcs), & w(dap) &\leq w(dbp), \\
w(dcq) &\leq w(daq), & w(csrbp) &\leq w(cqap), \\
w(qapbr) &\leq w(qcsr), & w(aqcs) &\leq w(apbrs).
\end{aligned}
\tag{3.2}$$

Adding (3.2) cancels every edge except br and yields $2w(br) \leq 0$. Thus Z_Δ is non-metrizable. The corresponding certificates for $Z_\star, Z_A, Z_B$ are the previously known K_4 -type certificates in [4, Appendix A and Fig. 22].

3.2 The neighborly obstruction N_{233}

Use signature $(2, 1, 1, 3, 1, 3)$ in the kernel-edge order 01, 02, 03, 12, 13, 23. Thus the edges are

$$04, 14, 02, 03, 15, 56, 26, 13, 27, 78, 38.$$

For each root x , let T_x be the spanning tree obtained by deleting the three edges in the following table, and let P_{xy} be the unique x - y path in T_x .

x	O_x	x	O_x	x	O_x
0	14, 56, 38	1	04, 26, 27	2	14, 15, 38
3	04, 26, 27	4	03, 56, 27	5	04, 02, 78
6	14, 03, 78	7	04, 03, 15	8	04, 03, 26

(3.3)

The paths supplied by T_x, T_y coincide for every pair x, y ; every edge incident with a root is retained. Hence (3.3) defines a neighborly consistent system. Six of its path inequalities are

$\{x, y\}$	P_{xy}	competitor
$\{0, 8\}$	0278	038
$\{0, 5\}$	0315	0265
$\{6, 8\}$	65138	6278
$\{4, 7\}$	41387	4027
$\{4, 6\}$	4026	4156
$\{5, 7\}$	5627	51387

(3.4)

Their integer edge-incidence vectors add to twice the vector of edge 13. Consequently any inducing weight would satisfy $2w(13) \leq 0$. The script `verify_neighborly_n233.py` reconstructs all nine trees and verifies consistency, neighborliness, and this identity without optimization or floating point arithmetic.

Shortening respectively the three long kernel edges of $P_4(2, 3, 3)$ gives a subdivision with at most two long edges, $P_4(2, 2, 3)$, or $P_4(2, 3, 2)$. All are positive cases proved below. Thus N_{233} is coordinate-minimal.

4 Length-shape reductions

Let $S(\ell) = \{ij : \ell_{ij} \geq 2\}$ be the long-edge support.

Proposition 4.1 (Ordinary shape reduction). *If $K_4(\ell)$ avoids the five graphs in (1.3), then, after relabelling the kernel, exactly one of the following holds:*

- (i) $|S(\ell)| \leq 2$;
- (ii) $S(\ell)$ is a triangle and its three lengths are $(2, 2, 2)$;
- (iii) $S(\ell)$ is a star and its three lengths are $(2, 2, 2)$;
- (iv) $S(\ell)$ is a three-edge path with consecutive lengths $(p, 2, r)$, or $(2, 3, 2)$.

Proof. Four long kernel edges contain either Z_A or Z_B , according as the two short complementary edges are adjacent or opposite. Hence $|S| \leq 3$. A three-edge subgraph of K_4 is a triangle, a star, or a path. A triangle or star with a length at least three contains Z_Δ or $Z_\star$, respectively.

For a path with lengths (p, q, r) , the endpoints of its middle edge are joined by three internally disjoint paths of lengths $q, p+1, r+1$. Since $p, r \geq 2$, all three theta arms have length at least three whenever $q \geq 3$. Avoidance of $\Theta_{3,3,4}$ then forces the three arm lengths $(q, p+1, r+1)$ to equal $(3, 3, 3)$, which is precisely $(p, q, r) = (2, 3, 2)$. Otherwise $q = 2$. $\square$

Proposition 4.2 (Neighborly shape reduction). *If a K_4 -subdivision contains no coordinate subdivision of the five graphs in (1.4), then it has one of the first three shapes in Proposition 4.1, or its long support is $P_4(p, q, r)$ with*

$$q = 2 \quad \text{or} \quad p = r = 2. \quad (4.1)$$

Proof. Avoiding the four Z -graphs gives the same support reduction, but imposes no theta restriction on a three-edge path. If $q \geq 3$ and one end length is at least three, the vector dominates $P_4(2, 3, 3)$, after reversing the kernel path if necessary. Thus avoidance of N_{233} leaves precisely (4.1). $\square$

5 Ferrers-potential tools

This section isolates the representation lemmas used in every infinite family. They are stated with strict comparisons so that no accidental tie changes a chosen route. The final inducing property remains non-strict.

5.1 Four corners

Let $I = [m]$, $J = [n]$, and let $I \times J = R_{00} \sqcup R_{01} \sqcup R_{10} \sqcup R_{11}$. Call the partition *corner closed* if each $R_{e\eta}$ is closed toward its named corner; for example, $(i, j) \in R_{00}$ and $i' \leq i, j' \leq j$ imply $(i', j') \in R_{00}$.

Lemma 5.1 (Four-corner representation). *Every corner-closed partition has strictly increasing labels $X_1 < \dots < X_m$, $Y_1 < \dots < Y_n$ and constants $C_{00}, C_{01}, C_{10}, C_{11}$ such that the color of (i, j) is the unique minimizer of*

$$C_{e\eta} + \sigma_e X_i + \sigma_\eta Y_j, \quad \sigma_0 = 1, \quad \sigma_1 = -1. \quad (5.1)$$

Proof. In a fixed row, the two left-closed colors cannot both occur, and neither can the two right-closed colors. Hence a row consists of one left color followed by one right color. Down the rows the left color changes at most once from 00 to 10, and the right color at most once from 01 to 11.

If the first change occurs no later than the second, the row types are 00/01, 10/01, 10/11. Put the three row bands into $(-\infty, 0)$, $(0, 1)$, $(1, \infty)$, do the same for columns, and within the central bands interleave labels so that the weakly monotone row boundary is given by $X_i < Y_j$. Then

$$X_i + Y_j, \quad X_i - Y_j, \quad -X_i + Y_j, \quad 2 - X_i - Y_j$$

have the required unique minima. If the change order is reversed, use the three bands

$$(-\infty, -1), \quad (-1, 0), \quad (0, \infty)$$

and constants $0, 0, 0, -2$; the central boundary is $X_i + Y_j < -1$. All labels may be perturbed away from the finitely many equality hyperplanes. Empty row bands, empty column bands, and constant row types require no separate argument: delete the empty band from the construction and choose the unused threshold in the adjacent open interval. Thus the proof includes all boundary cases $R_{e\eta} = \emptyset$ and $m = 1$ or $n = 1$. $\square$

For two suspended chains, subpath consistency makes the table of selected single-bridge routes corner closed: a route entering both chains from their left endpoints forces the same route on every southwest subpair, and similarly at the other corners. Chain increments $w(a_{i-1}a_i) = (X_i - X_{i-1})/2$ translate (5.1) directly into path lengths.

5.2 Difference relations with a prescribed margin

A relation $F \subseteq \{0, \dots, p\} \times \{p+1, \dots, m\}$ is *expanding* if

$$(i, j) \in F, \quad i' \leq i, \quad j' \geq j \quad \implies \quad (i', j') \in F. \quad (5.2)$$

Lemma 5.2 (Fixed-margin outward-safe extension). *Suppose every row $i \leq p$ of an expanding relation F is nonempty. Assume a finite family of already strict auxiliary comparisons remains true when the active prefix $X_0, \dots, X_p$, with its internal order fixed, is moved sufficiently far to the left. If a prescribed prefix cut contains the active prefix, then for every sufficiently large $A > 0$ there are increasing labels and T preserving the auxiliary comparisons and the cut such that*

$$(i, j) \in F \iff X_j - X_i > T, \quad T - \frac{X_m - X_0}{2} = A. \quad (5.3)$$

A symmetric statement holds for an active suffix moved to the right.

Proof. Keep $X_{p+1} < \dots < X_m$ fixed and let $r_i = \min\{j : (i, j) \in F\}$. Then $r_0 \leq \dots \leq r_p$. Choose strictly increasing H_i with

$$X_{r_i-1} < H_i < X_{r_i}, \quad (5.4)$$

interpreting the left endpoint as $-\infty$ when $r_i = p + 1$. Put

$$T = 2A + X_m - H_0, \quad X_i = H_i - T \quad (0 \leq i \leq p). \quad (5.5)$$

For large A , the prefix lies in the outward-safe region and does not cross the prescribed cut. Moreover $X_j - X_i > T \iff X_j > H_i \iff j \geq r_i$, while differences on either side of the cut are smaller than T . Finally

$$T - \frac{X_m - X_0}{2} = \frac{T - X_m + H_0}{2} = A.$$

□

The following observation supplies the hypothesis that was implicit in early versions of the argument.

Lemma 5.3 (Consistency gives outward safety). *Consider either two suspended kernel chains joined at their endpoints, or the split-bridge graph defined in Section 6.2. If a same-chain selected path detours outside the chain, then the active left endpoints occur only in cross-chain routes whose affine expression has coefficient $+X_i$, and the active right endpoints only in routes with coefficient $-X_i$. For the second chain the analogous coefficients are $+Y_j$ and $-Y_j$. In the five-route notation of Section 6.2, the allowed color sets are*

$$\begin{array}{ll} X \text{ moved left} & \left| \begin{array}{l} \{00, 01\} \\ \{00, 10, W\} \end{array} \right. \\ Y \text{ moved left} & \left| \begin{array}{l} \{10, 11, W\} \\ \{01, 11\} \end{array} \right. \end{array} \quad (5.6)$$

Consequently the relevant active side is outward safe in the sense of Lemma 5.2.

Proof. Let $P_{a_i a_j}$, $i < j$, be a detour. Its two portions from the detour route to a_i, a_j force, by consistency, the entry directions at its two ends. More explicitly, if i is an active left endpoint, any cross route that first follows the chain from a_i toward its right endpoint passes through the matching active right endpoint a_j ; it would therefore contain the direct a_i - a_j chain segment as a subpath, contradicting $P_{a_i a_j}$. Thus such a cross route must leave through the left endpoint. The same argument with the orientation reversed applies to an active right endpoint. Inspecting the endpoint used by each of the four single-bridge routes gives $\{00, 01\}$ on the left and $\{10, 11\}$ on the right. In the split bridge, the complementary route W enters the first chain at its right endpoint and the second at its left, which adds it exactly to the two sets shown in (5.6). The argument for the second chain is identical. Within one listed set all affine expressions have the same coefficient in the moving coordinate, so moving the block outward does not change their relative order; expressions of the opposite coefficient are excluded by the subpath argument. This is precisely outward safety. □

5.3 Two prescribed Ferrers gaps

Lemma 5.4 (Additive double-gap lemma). *Let $F \subseteq [m] \times [n]$ be a down-set. There are increasing labels x_i, y_j with $(i, j) \in F \iff x_i + y_j < 0$, and prescribed gaps*

$$x_k < a < x_{k+1}, \quad y_l < b < y_{l+1}, \quad (5.7)$$

if and only if

$$\begin{array}{l|l} a + b < 0 & (k, l) \in F, \\ a + b = 0 & (k, l) \in F \text{ and } (k+1, l+1) \notin F, \\ a + b > 0 & (k+1, l+1) \notin F. \end{array} \quad (5.8)$$

All labels may additionally be confined to any common open interval containing a and $-b$.

Proof. Set $z_j = -y_j$. The down-set determines an acyclic total comparison system by $x_i < z_j$ for members and $z_j < x_i$ for nonmembers, together with the two internal orders. Insert a between x_k, x_{k+1} and $-b$ between z_{l+1}, z_l . Their required relative order gives exactly (5.8). A finite strict order embeds into any prescribed open interval. $\square$

Lemma 5.5 (Difference double-gap lemma). *Let $Q \subseteq [m] \times [n]$ be southeast closed. There are increasing labels X_i, z_j satisfying*

$$(i, j) \in Q \iff X_i > z_j$$

and prescribed gaps

$$X_k < a < X_{k+1}, \quad z_l < c < z_{l+1} \quad (5.9)$$

under exactly the following conditions:

$$\begin{array}{l|l} a < c & (k, l+1) \notin Q, \\ a = c & (k, l+1) \notin Q \text{ and } (k+1, l) \in Q, \\ a > c & (k+1, l) \in Q. \end{array} \quad (5.10)$$

Proof. Order the symbols X_i, z_j by their two internal orders and, for every cell, by $z_j < X_i$ if the cell belongs to Q and by $X_i < z_j$ otherwise. Southeast closure is precisely the condition that these comparisons are acyclic; indeed every X_i, z_j pair is already comparable. If $a < c$, inserting the two new symbols into the gaps in (5.9) can fail only when $z_{l+1} < X_k$, which is $(k, l+1) \in Q$. If $a > c$, the symmetric condition is $z_l < X_{k+1}$, namely $(k+1, l) \in Q$. For equality the two open gaps must intersect, requiring both comparisons. A linear extension of the resulting finite order supplies the labels. $\square$

6 The ordinary positive families

6.1 At most two long kernel edges

We first isolate the point that is most easily hidden by a short four-corner argument: when two long kernel edges are adjacent, the two prefixes selected from the fourth kernel vertex and the midpoint inequalities for the four short edges must be realized by the same coordinates.

Lemma 6.1 (Adjacent four-route normal form). *Let*

$$A : z = a_0, \dots, a_m = u, \quad B : z = b_0, \dots, b_n = v$$

be the two possibly subdivided adjacent kernel edges, and let the other four edges be zx, ux, vx, uv . In a neighborly consistent path system:

(i) every selected a_i - b_j path is exactly one of

$$\begin{array}{l|l} S & a_iAzBb_j \\ T & a_iAu - vBb_j \\ U & a_iAz - x - vBb_j \\ V & a_iAu - x - zBb_j; \end{array} \quad (6.1)$$

(ii) the sets

$$D_A = \{i : P_{xa_i} = xzAa_i\}, \quad D_B = \{j : P_{xb_j} = xzBb_j\} \quad (6.2)$$

are prefixes, with 0 in and the last index out;

(iii) there are strictly increasing labels ξ_i, η_j and numbers $\Delta < h < 0$ such that (6.2) is given by $\xi_i < h, \eta_j < h$, and the selected route in (6.1) is the unique minimum of

$$L_S = \xi_i + \eta_j, \quad L_U = \xi_i - \eta_j, \quad L_V = -\xi_i + \eta_j, \quad L_T = 2\Delta - \xi_i - \eta_j. \quad (6.3)$$

Proof. A selected path is chordless. Once its initial and terminal chain segments are removed, an induced path in the four-vertex kernel with the two suspended chains can use only one of the four arcs in (6.1): the route through $u - x - v$ has chord uv , the route through $z - x - u - v$ has chord xv , and every longer kernel walk either repeats a vertex or contains one of these chords. This proves (i).

The only chordless paths from x to a_i enter A at z or at u . If the first choice occurs at i , consistency gives it for every smaller index, proving that D_A is a prefix; neighborliness gives its two endpoint conditions. The argument for D_B is identical. Taking the x -to-chain subpaths of (6.1) gives

$$U(i, j) \Rightarrow i \in D_A, j \notin D_B, \quad V(i, j) \Rightarrow i \notin D_A, j \in D_B. \quad (6.4)$$

Thus a row cannot contain both U and V . Corner closure follows by taking subpaths. In addition, every row containing U has the same S/U cut: if the same column contained U in one such row and S in another, the two selected paths would prescribe different z - b_j subpaths. The analogous assertion holds for V , and the column statements are symmetric.

Delete these constant-color outer rows and columns. Only S, T remain in the central rectangle, and the S -cells form a product-order down-set. Write this down-set as

$$S(i, j) \iff r_i + s_j < 0 \quad (6.5)$$

with both label sequences strictly increasing. The cuts D_A, D_B leave open gaps in the two sequences. Replacing r_i by $r_i + t$ and s_j by $s_j - t$ preserves every sum, and an appropriate t makes the two gaps overlap. Choose a common point g in their overlap. After a positive scaling, put

$$\xi_i = \Delta/2 + \varepsilon r_i, \quad \eta_j = \Delta/2 + \varepsilon s_j, \quad h = \Delta/2 + \varepsilon g \quad (6.6)$$

on the central rows and columns. Taking ε small places them in $(\Delta, 0)$. Place the outer U -rows and V -columns below Δ , and the opposite outer rows and columns above 0, preserving their internal order. Direct comparison in (6.3) gives

	$\eta < \Delta$	$\Delta < \eta < 0$	$\eta > 0$
$\xi < \Delta$	S	S	U
$\Delta < \xi < 0$	S	S/T	T
$\xi > 0$	V	T	T

(6.7)

This recovers all outer rectangles, the central staircase, and both prefixes, proving (iii). Empty outer or central bands are handled by omitting the band and placing its unused threshold in the adjacent open gap. $\square$

Lemma 6.2 (Adjacent midpoint normalization). *The representation in Lemma 6.1 may be chosen simultaneously with both same-chain detour relations so that, for*

$$\begin{aligned}\mu_A &= (\xi_0 + \xi_m)/2, & \mu_B &= (\eta_0 + \eta_n)/2, \\ \mu_A &< h/2, & \mu_B &< h/2, & \mu_A + \mu_B &< \Delta.\end{aligned}\tag{6.8}$$

If the selected z - u path detours through x , its difference threshold can be prescribed as $-\xi_0$; if it detours through $B + vu$, it can be prescribed as $\Delta - \xi_0 - \eta_0$. The symmetric statement holds for B .

Proof. Every same-chain detour relation is expanding. All its left endpoints precede all its right endpoints, and all detours use the same exterior endpoint path; otherwise two detours would prescribe different paths between the kernel endpoints. By Lemma 5.3, its active left block can be moved left and its active right block moved right without changing the cross-route table or either prefix.

We need the following quantitative form of that move. Suppose the active left indices are $0, \dots, p$, a desired threshold is $\tau = C - \xi_0$, and r_j is the largest active left index paired with a right index j . Choose $\lambda_0 > \dots > \lambda_p > 0$ and set

$$\xi_i(t) = \bar{\xi}_i - \lambda_i t \quad (0 \leq i \leq p).\tag{6.9}$$

Then $\tau(t) = C - \bar{\xi}_0 + \lambda_0 t \rightarrow \infty$. For a nonempty right column choose

$$H_j(t) = -\gamma_j t + O(1), \quad \lambda_{r_j} > \gamma_j > \lambda_{r_j+1},\tag{6.10}$$

with the evident end-interval convention when $r_j = p$, and put $\xi_j = H_j + \tau$. Repeated values of r_j are separated by their bounded terms. Hence

$$(i, j) \text{ detours} \iff \xi_j - \xi_i > \tau, \quad \xi_j \rightarrow +\infty, \quad H_m \rightarrow -\infty.\tag{6.11}$$

An empty right column retains its old label; its insertion value moves with ξ_0 , so its cross color is unchanged. Formula (6.11) is the required double-scale extension.

There are three endpoint configurations. If neither endpoint path uses the other chain, each is direct or uses x . In the direct case the boundary cell gives $\xi_m < 0$ while $\xi_0 < h$, and moving ξ_0 left makes μ_A arbitrarily negative. In the x -detour case take $\tau_A = -\xi_0$. Then (6.11) gives $2\mu_A = H_m$, again arbitrarily negative. Treat B independently; at an intersection of the two moving blocks the color is S , which only becomes more strongly minimal when both coordinates move left.

Next suppose the z - u path uses $B + vu$. Consistency forces the z - v path to be the direct chain B , and excludes U, V from the whole cross table. The active A -rows are therefore all S and are left-outward safe. Set $\tau_A = \Delta - \xi_0 - \eta_0$. Formula (6.11) yields

$$2\mu_A = H_m + \Delta - \eta_0.\tag{6.12}$$

The direct B -endpoint path gives $2\mu_B = \eta_0 + \eta_n < h$. Since H_m can be arbitrarily negative, choose it so that both $2\mu_A < h$ and $2(\mu_A + \mu_B) < 2\Delta$. The case in which B uses $A + uv$ is symmetric. Two endpoint paths cannot both use the other chain, because each would prescribe a different kernel-endpoint subpath. These cases exhaust all possibilities and prove (6.8) with the asserted thresholds. $\square$

Theorem 6.3 (Two-chain theorem). *Every K_4 -subdivision with at most two long kernel edges is metrizable.*

Proof. It suffices first to treat neighborly systems. If the two suspended chains are opposite, every chordless cross-chain selected path uses exactly one of the four endpoint bridges. Consistency makes the resulting color table corner closed, so Lemma 5.1 realizes all cross-chain choices. A same-chain detour relation is expanding, all its intervals meet, and all detours use a single endpoint of the other chain. Apply Lemmas 5.2 and 5.3 on the side outside the active four-corner staircase. If $M_A = (X_m - X_0)/2$, a detour threshold T_A is realized by prescribing the corresponding bridge-column sum to be $T_A - M_A$; unused column sums are taken larger. The second chain is symmetric.

The alternating difference of the four bridge constants is the only invariant of the four-corner table under row and column shifts. After fixing it, any specified positive row and column sums can be met: with one row sum B , one column sum A , and alternating difference d , take

$$q_{00} = x, \quad q_{01} = B - x, \quad q_{10} = A - x, \quad q_{11} = A + B + d - 3x,$$

where $x > 0$ is small and A, B are made large by the fixed-margin extension. Fewer specified sums are easier. Thus every chordless selected path is shortest, and Lemma 2.3 completes the opposite case.

When the long edges are adjacent, apply Lemmas 6.1 and 6.2. Center the labels by $X_i = \xi_i - \mu_A$, $Y_j = \eta_j - \mu_B$, give each chain edge half of the corresponding label increment, and set

$$w(zx) = -h/2, \quad w(ux) = h/2 - \mu_A, \quad w(vx) = h/2 - \mu_B, \quad w(uv) = \Delta - \mu_A - \mu_B. \quad (6.13)$$

All four weights are positive by (6.8). After deletion of a common term, twice the four cross-route lengths are exactly (6.3); the two x -entry comparisons are exactly the cuts in (6.2), and the prescribed thresholds in Lemma 6.2 realize every same-chain detour. Hence all chordless selected paths are shortest, and Lemma 2.3 completes the adjacent case.

Finally suppose the system is not neighborly. By Lemma 2.2, delete an unused edge. In a subdivision of K_4 , every nontrivial block after such a deletion is a theta graph (the two broken pieces of the deleted kernel path are pendant) or a cycle. It avoids $\Theta_{3,3,4}$ because the original graph does. Hence it is metrizable by (1.1), and Lemma 2.2 restores the edge. $\square$

6.2 The split-bridge family $P_4(p, 2, r)$

Let

$$A : a_0, \dots, a_p, \quad B : b_0, \dots, b_r$$

be disjoint chains. Add the three edges a_0b_0, a_0b_r, a_pb_r and the two-edge path $a_p - s - b_0$. The result, denoted $W_{p,r}$, is $P_4(p, 2, r)$.

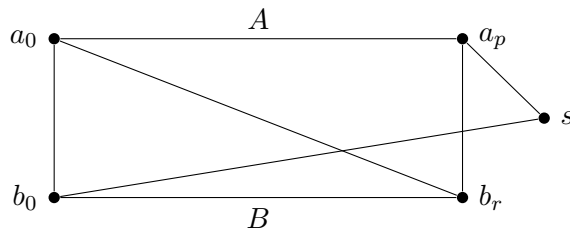


Figure 2: The split-bridge graph $W_{p,r} = P_4(p, 2, r)$. The path through s is the Q -route; the complementary arc $a_p - b_r - a_0 - b_0$ is the fifth route W .

Besides the four single-connection routes, there is one additional chordless cross route

$$W(i, j) = a_i A a_p - b_r - a_0 - b_0 B b_j. \quad (6.14)$$

The route using $a_p - s - b_0$ is denoted by Q . A selected Q -route and a selected W -route cannot coexist, because they prescribe different a_p - b_0 subpaths. Hence every neighborly system lies in a global Q -mode, a global W -mode, or an N -mode using neither route.

Write q_{00}, q_{01}, q_{11} for the three ordinary bridge weights and $q_{10} = w(a_p s) + w(s b_0)$. Put

$$\delta = w(a_p s) - w(s b_0), \quad \mu_A = \frac{X_0 + X_p}{2}, \quad \mu_B = \frac{Y_0 + Y_r}{2}. \quad (6.15)$$

The three chordless routes from s to a point on A are

$$R_A = s - a_p A a_i, \quad L_A = s - b_0 - a_0 A a_i, \quad D_A = s - a_p - b_r - a_0 A a_i,$$

and symmetrically on B . Consistency makes the non- R_A indices a prefix using only L_A or only D_A ; on B , the L_B indices form a prefix followed only by R_B or only by D_B . The two diagonal modes cannot coexist. Direct comparison gives

$$\begin{aligned} L_A < R_A &\iff X_i < \mu_A + \delta - q_{00}, & D_A < R_A &\iff X_i < \mu_A - q_{01} - q_{11}, \\ L_B < R_B &\iff Y_j < \mu_B + \delta + q_{11}, & L_B < D_B &\iff Y_j < \mu_B + q_{00} + q_{01}. \end{aligned} \quad (6.16)$$

The split weights are positive precisely when $-q_{10} < \delta < q_{10}$.

Lemma 6.4 (Five-route exhaustion and entrance intervals). *For a neighborly consistent system on $W_{p,r}$, the chordless paths between the two chains consist precisely of the four routes that use one of*

$$a_0 b_0, \quad a_0 b_r, \quad a_p s b_0, \quad a_p b_r$$

and, when its end segments do not create a chord, the complementary route $W(i, j)$ in (6.14). The short route $Q = a_p s b_0$ and W cannot both be selected anywhere. Thus every system belongs uniquely to a Q -mode, a W -mode, or an N -mode using neither.

For paths rooted at s , the nontrivial possibilities are exactly

$$\begin{array}{l|l} R_A & s - a_p A a_i \\ L_A & s - b_0 - a_0 A a_i \\ D_A & s - a_p - b_r - a_0 A a_i \end{array} \quad \begin{array}{l|l} L_B & s - b_0 B b_j \\ R_B & s - a_p - b_r B b_j \\ D_B & s - b_0 - a_0 - b_r B b_j. \end{array} \quad (6.17)$$

On A , the non- R_A indices form a prefix using only L_A or only D_A ; on B , the L_B indices form a prefix followed only by R_B or only by D_B . The two diagonal modes cannot coexist.

Let E_A be the non- R_A prefix and E_B the L_B prefix. If $I_A = (X_k - \mu_A, X_{k+1} - \mu_A)$ and $I_B = (Y_l - \mu_B, Y_{l+1} - \mu_B)$ are their nontrivial cut gaps, put

$$M_A = (X_p - X_0)/2, \quad M_B = (Y_r - Y_0)/2, \quad \kappa_A = q_{00} - q_{01} - q_{11}, \quad \kappa_B = q_{00} + q_{01} - q_{11}. \quad (6.18)$$

The admissible values of δ are exactly

$$\mathcal{J}_A = \begin{cases} (q_{00} + I_A) \cap (\kappa_A, \infty), & A : L, \\ (-\infty, \kappa_A), & A : D, -q_{01} - q_{11} \in I_A, \\ (-\infty, q_{00} - M_A), & A : R, q_{01} + q_{11} > M_A, \end{cases} \quad (6.19)$$

and

$$\mathcal{J}_B = \begin{cases} (-q_{11} + I_B) \cap (-\infty, \kappa_B), & B : R, \\ (\kappa_B, \infty), & B : D, q_{00} + q_{01} \in I_B, \\ (M_B - q_{11}, \infty), & B : L, q_{00} + q_{01} > M_B. \end{cases} \quad (6.20)$$

Consequently all s -root paths and both split-edge positivities are realized if and only if

$$\mathcal{J}_A \cap \mathcal{J}_B \cap (-q_{10}, q_{10}) \neq \emptyset. \quad (6.21)$$

Proof. Remove the initial and terminal chain segments from an induced cross path. In the five-vertex kernel obtained by subdividing the fourth bridge, an induced path either crosses exactly one bridge or follows the complementary arc of the five-cycle. Every other kernel walk repeats a vertex or contains one of the three ordinary bridges as a chord. The complementary arc is precisely $W(i, j)$. Every Q -path contains $P_{a_p b_0} = a_p s b_0$, whereas every W -path contains $P_{a_p b_0} = a_p b_r a_0 b_0$; consistency proves mode separation.

After the first edge from s , the same induced-path inspection gives the six routes in (6.17), including the boundary restrictions caused by the chords $s a_p$ and $a_p a_{p-1}$. Taking subpaths shows that all non-right entries on A form a prefix. Two entries of that prefix cannot use both L_A and D_A , since they prescribe different s - a_0 subpaths. The argument on B is symmetric. A D_A -path prescribes $P_{s b_r} = s a_p b_r$, while a D_B -path prescribes $P_{s b_r} = s b_0 a_0 b_r$; hence they cannot coexist. Taking the two subpaths of a $Q(i, j)$ -route also gives

$$Q(i, j) \implies i \notin E_A, \quad j \in E_B. \quad (6.22)$$

Let $q_{10} = \alpha + \beta$ and $\delta = \alpha - \beta$. With the centered chain coordinates, the six route lengths are

$$\begin{array}{l|l} R_A & \alpha + (X_p - X_i)/2 \\ L_A & \beta + q_{00} + (X_i - X_0)/2 \\ D_A & \alpha + q_{11} + q_{01} + (X_i - X_0)/2 \end{array} \quad \begin{array}{l|l} L_B & \beta + (Y_j - Y_0)/2 \\ R_B & \alpha + q_{11} + (Y_r - Y_j)/2 \\ D_B & \beta + q_{00} + q_{01} + (Y_r - Y_j)/2. \end{array} \quad (6.23)$$

Pairwise comparison of the three entries in each column gives (6.19) and (6.20); for example

$$L_A < R_A \iff X_i < \mu_A + \delta - q_{00}, \quad L_A < D_A \iff \delta > \kappa_A.$$

The all-right and all-left cases require the indicated endpoint inequalities. Finally $\alpha, \beta > 0$ is equivalent to $-q_{10} < \delta < q_{10}$, proving (6.21). $\square$

Lemma 6.5 (Mode normal forms). *The cross-route table in each global mode has the following exact form.*

- (a) In W -mode, the W -cells and 01-cells are two opposite rectangles. All remaining cells are 00/11, and the 00-cells form a product-order down-set.
- (b) In Q -mode, put $d = q_{00} + q_{11} - q_{01} - q_{10}$. If $d \leq 0$, the Q - and 01-cells are opposite rectangles with a central additive 00/11 staircase. If $d > 0$, the central rectangle is a difference staircase

$$Q(i, j) \iff X_i - Y_j > q_{10} - q_{01}. \quad (6.24)$$

- (c) In N -mode, the 01-cells form a rectangle and all remaining cells form an additive 00/11 staircase.

Proof. In W -mode, let F_A, F_B be the two same-chain detour relations and put $R_A = \{i : (0, i) \in F_A\}$, $L_B = \{j : (j, r) \in F_B\}$. A $W(i, j)$ path has both detour subpaths, so its cell lies in $R_A \times L_B$. Conversely, in that product the colors 00, 01 contradict the A -detour, the colors 01, 11 contradict the B -detour, and Q is excluded by the global mode. Thus

$$c(i, j) = W \iff (i, j) \in R_A \times L_B. \quad (6.25)$$

If R_{01} and C_{01} are the rows and columns in which 01 occurs, corner closure makes them a prefix and a suffix. If a cell in their product were 00 or 11, it would contain a subpath conflicting with an 01-cell in the same column or row. It cannot be W by (6.25), so the 01-cells are exactly $R_{01} \times C_{01}$. Corner closure now gives the central 00/11 down-set.

For Q -mode, compare the four affine expressions and put

$$r_0 = q_{10} - q_{00}, \quad r_1 = q_{11} - q_{01}, \quad c_0 = q_{01} - q_{00}, \quad c_1 = q_{11} - q_{10}.$$

Then $r_1 - r_0 = c_1 - c_0 = d$, and Q is minimal exactly when

$$X_i > r_0, \quad Y_j < c_1, \quad X_i - Y_j > q_{10} - q_{01}. \quad (6.26)$$

For $d \leq 0$ the third inequality follows from the first two, giving the rectangle; the symmetric calculation gives the opposite 01-rectangle. For $d > 0$ their overlap is the central difference relation (6.24).

In N -mode the same row-and-column subpath argument used for 01 in the first paragraph gives a rectangle. Cells southwest and northeast of it are forced to 00 and 11, respectively, and corner closure orders the remaining two-color cells. This proves all three normal forms. $\square$

Lemma 6.6 (Unified split-bridge scalar feasibility). *Let $\mathcal{P}$ be a neighborly consistent path system on $W_{p,r}$, and encode its cross-chain routes, its paths rooted at s , and its two same-chain detour relations as in Lemmas 6.4 and 6.5. There are strictly increasing rational labels*

$$X_0 < \dots < X_p, \quad Y_0 < \dots < Y_r,$$

rational detour thresholds T_A, T_B , positive rational bridge parameters

$$q_{00}, q_{01}, q_{11}, \alpha, \beta, \quad q_{10} = \alpha + \beta, \quad \delta = \alpha - \beta, \quad (6.27)$$

with the following simultaneous properties.

- (i) *The selected entry in every cross-chain cell is strictly shorter than each of the other chordless cross routes available in that cell.*
- (ii) *The six possible s -root routes in (6.17) have exactly the prescribed entrance cuts. Equivalently,*

$$\delta \in \mathcal{J}_A \cap \mathcal{J}_B \cap (-q_{10}, q_{10}). \quad (6.28)$$

In particular, both edges incident with s have positive weights $\alpha = (q_{10} + \delta)/2$ and $\beta = (q_{10} - \delta)/2$.

- (iii) *Every same-chain detour is represented by its prescribed expanding difference relation, using the same labels X, Y ; every unavailable member of the pair Q, W is strictly longer than the selected cross route.*
- (iv) *Any finite family of strict comparisons already imposed by the relevant Ferrers normal form remains strict.*

All choices are obtained by rational order insertions and explicit rational affine transformations.

Proof. We separate the order-theoretic compatibility from the scalar calibration. The central relation in a mode normal form is either an additive down-set or a difference staircase. Suppose first that a down-set F must carry two entrance gaps, after rows k and columns l . Since F is a down-set, precisely one of the following occurs:

$$(k+1, l+1) \in F; \quad (k, l) \in F, (k+1, l+1) \notin F; \quad (k, l) \notin F. \quad (6.29)$$

Choose the sum of the two inserted gap values to be respectively negative, zero, or positive. These are exactly the three alternatives in (5.8), so Lemma 5.4 inserts both cuts simultaneously. For a difference staircase, the required orientation of the two gaps is recorded by the subpaths through a_p and b_0 : implication (6.22) gives $(k, l+1) \notin Q$ in the $a < c$ orientation, whereas a diagonal entrance gives $(k+1, l) \in Q$ in the $a > c$ orientation. If both statements hold, the two open gaps overlap and we use $a = c$. Thus the alternatives are exactly (5.10), and Lemma 5.5 applies. This verifies the two-gap interface without a continuity or generic-position assumption.

We now calibrate the three global modes. Throughout, centering a chain means subtracting its midpoint; this changes only the constant bridge terms. Strict order insertions may be taken rational.

The W-mode. Let F_A, F_B be the same-chain detour relations and put

$$R_A = \{i : (0, i) \in F_A\}, \quad L_B = \{j : (j, r) \in F_B\}.$$

By Lemma 6.5, the W -cells are exactly $R_A \times L_B$, the 01-cells form the opposite rectangle, and the remaining 00/11 cells form an additive down-set. Put $M_A = (X_p - X_0)/2$, $M_B = (Y_r - Y_0)/2$, $A_0 = T_A - M_A$, and $B_0 = T_B - M_B$. The fixed-margin extension lets us make $A_0, B_0 > 0$; if one detour relation is empty, its unused threshold is chosen directly beyond the corresponding half-span. For $0 < q < \min\{A_0, B_0\}$, set

$$q_{01} = q, \quad q_{11} = A_0 - q, \quad q_{00} = B_0 - q, \quad \hat{q}_{10} = A_0 + B_0 - q. \quad (6.30)$$

With $u_i = X_i - A_0$, $v_j = Y_j + B_0$, the four relevant expressions, after deletion of a common term, are

route	00	01	11	W	(6.31)
expression	$u_i + v_j$	$4q + u_i - v_j$	$-u_i - v_j$	$-u_i + v_j$	

The two rectangles and the central additive relation are therefore realized strictly. The alternatives in (6.29) realize both entrance cuts. Choose the resulting δ rational and then take $q_{10} > \max\{\hat{q}_{10}, |\delta|\}$. This preserves the table, makes $\alpha, \beta > 0$, and makes every unused Q -route longer.

The Q-mode. Use the independent Hadamard coordinates

$$\begin{aligned} d &= q_{00} + q_{11} - q_{01} - q_{10}, \\ C &= (q_{01} + q_{11}) - (q_{00} + q_{10}), \\ R &= (q_{10} + q_{11}) - (q_{00} + q_{01}), \\ \Lambda &= q_{00} + q_{01} + q_{10} + q_{11}. \end{aligned} \quad (6.32)$$

Their inverse is

$$\begin{aligned} q_{00} &= (\Lambda + d - C - R)/4, & q_{01} &= (\Lambda - d + C - R)/4, \\ q_{10} &= (\Lambda - d - C + R)/4, & q_{11} &= (\Lambda + d + C + R)/4. \end{aligned} \quad (6.33)$$

The separation of roles is summarized below.

parameter	comparison controlled
d	additive versus difference central staircase
C	the A -side diagonal interval
R	the B -side diagonal interval
Λ	common bridge positivity and ordinary-interval width
δ	balance of the two split edges and the entrance cuts.

Thus d, C, R control respectively the central staircase, the A -side diagonal direction, and the B -side diagonal direction, while increasing Λ makes every bridge positive without changing those three data. For $d \leq 0$, the central 00/11 down-set is handled by (6.29). For $d > 0$, the central Q /01 staircase is handled by the difference-gap compatibility proved in the first paragraph.

The remaining scalar condition is the open interval (6.28). From (6.19) and (6.20), its three nondegenerate possibilities are

$$\begin{aligned} (L_A, R_B) & \left| \begin{array}{l} \max\{-q_{10}, -q_{10} - C\} < \delta < \min\{q_{10}, q_{10} - R\}, \\ -q_{10} < \delta < -q_{10} - C, \\ q_{10} - R < \delta < q_{10}. \end{array} \right. \end{aligned} \quad (6.34)$$

The A -diagonal interval is nonempty exactly when $C < 0$, and the B -diagonal interval is nonempty exactly when $R > 0$. We impose these signs when the corresponding entrance is

diagonal. They are independent Hadamard coordinates, and the two diagonal entrances cannot coexist by Lemma 6.4. The last two intervals then have widths $-C$ and R , respectively, while the first interval is nonempty once Λ , and therefore q_{10} , is large. There is no hidden dependence between the gap orientation and these signs. To see this directly, write $t = q_{10}$, $P = t + d$, choose $0 < \varepsilon < t/3$, and use

A -gap	B -gap	(P, δ)
before	inside	$(0, -\varepsilon)$
before	after	$(2\varepsilon, 0)$
inside	inside	$(-2\varepsilon, 0)$
inside	after	$(0, \varepsilon)$.

(6.35)

If $a = q_{00}$, $b = q_{11}$, and $q_{01} = a + b - P$, then $C = 2b - (P + t)$ and $R = P + t - 2a$. Choosing a, b in rational open intervals selects the required signs independently while retaining $a, b, q_{01} > 0$.

If an entrance prefix is empty or full, apply the common lift

$$q_{\varepsilon\eta} \mapsto q_{\varepsilon\eta} + \lambda \quad (6.36)$$

and, according to the retained ordinary cut, send δ to $\delta - \lambda$, $\delta + \lambda$, or leave it fixed. Every cross-route expression receives the same increment. The missing midpoint margin and the unused- W margin $q_{00} + q_{01} + q_{11} - q_{10}$ grow by 2λ . The fixed-margin outward-safe extension moves the active labels without changing the table and matches any detour threshold that must follow a lifted row or column sum.

The N -mode. Set

$$q_{00} = q_{11} = u, \quad q_{01} = v, \quad \rho = u - v. \quad (6.37)$$

Outside the 01-rectangle, the normal form becomes

$$01 \text{ minimal} \iff X_i < \rho, Y_j > -\rho, \quad 00 \text{ minimal} \iff X_i + Y_j < 0. \quad (6.38)$$

After a positive rational rescaling of the labels, the four relative positions of the ordinary entrance cuts are represented by $(u, v, \delta) = (1, 3, -2), (1, 1, 0), (1, 3, 0), (1, 3, 2)$. In the sole two-gap case, choose a sufficiently small rational perturbation of δ with the sign selected by (6.29). A single diagonal entrance is obtained by the corresponding choice $v > u$ or $v < u$; empty and full prefixes are handled by the common lift (6.36). Now take rational $q_{10} > |\delta|$ larger than the total weight of the old graph. Then no old-vertex path through s is competitive. Cell by cell, the difference between the unused W -route and the selected expression is one of

$$2(u + v - X_i), \quad 2(2u - X_i + Y_j), \quad 2(u + v + Y_j), \quad (6.39)$$

which is positive by the rectangle boundary and the corresponding detour condition.

Finally apply Lemmas 5.2 and 5.3 independently on every active chain side. Outward safety says that all expressions using an active label have the same coefficient, so these moves preserve the strict cross-route and entrance comparisons. The fixed-margin formula realizes the same-chain difference thresholds exactly. All restrictions are strict linear inequalities with rational coefficients; the order insertions and the free parameters above can therefore be chosen rational while avoiding the finitely many equality hyperplanes. This proves (i)–(iv) simultaneously. $\square$

Theorem 6.7 (Split-bridge realization). *For all $p, r \geq 2$, every neighborly consistent path system on $W_{p,r} = P_4(p, 2, r)$ is induced by positive edge weights.*

Proof. Apply Lemma 6.6. Give the two chain edges the positive weights

$$w(a_{i-1}a_i) = \frac{X_i - X_{i-1}}{2}, \quad w(b_{j-1}b_j) = \frac{Y_j - Y_{j-1}}{2},$$

give the three ordinary bridges weights q_{00}, q_{01}, q_{11} , and put $w(a_p s) = \alpha$, $w(s b_0) = \beta$. Parts (i)–(iii) of the lemma compare every chordless cross-chain, s -root, and same-chain competitor. Every remaining competitor has a chord and reduces to these comparisons by Lemma 2.3. Hence every selected path is shortest under a positive rational weighting. $\square$

6.3 The two irreducible finite ordinary cores

Proposition 6.8 (Exact finite-core theorem). *The two graphs*

$$K_4(2, 2, 1, 2, 1, 1) \quad \text{and} \quad K_4(2, 2, 2, 1, 1, 1) \quad (6.40)$$

are metrizable.

Proof. For neighborly systems, the complete rooted-tree enumeration described in Section 8 gives respectively 510 and 542 systems. The archive contains a positive rational inducing vector for each of these 1,052 systems. The independent verifier re-enumerates the systems, checks that no record is missing or duplicated, and tests every selected path against every simple competitor using exact fractions. Although the archive also contains 3,302 records for $K_4(2, 1, 1, 3, 1, 2) = P_4(2, 3, 2)$, those records are redundant for the proof: that graph is handled symbolically as H_3 in Corollaries 7.3 and 7.4.

For a non-neighborly system delete an unused edge by Lemma 2.2. The only nontrivial block is a theta subdivision or a cycle. For the triangle core, breaking the corridors in the order 01, 02, 03, 12, 13, 23 gives theta-arm triples

$$(1, 3, 3), (1, 3, 3), (2, 2, 4), (1, 3, 3), (2, 2, 4), (2, 2, 4),$$

whereas the corresponding list for the star core is

$$(1, 2, 4), (1, 2, 4), (1, 2, 4), (2, 3, 3), (2, 3, 3), (2, 3, 3).$$

Thus every theta block has an arm of length at most two and is metrizable by (1.1); cycle and tree blocks are metrizable as well. Restore the unused edge with sufficiently large weight. $\square$

7 The neighborly long-ear family

It remains, by Proposition 4.2, to treat

$$H_q = P_4(2, q, 2) \quad (q \geq 2). \quad (7.1)$$

Write its long chain as $C : x_0 = s, x_1, \dots, x_q = t$. Removing the internal vertices of C leaves a six-vertex graph J : two length-three s - t paths together with a chord between one internal vertex of each. Let $K = J + st$.

Fix a neighborly consistent system on H_q . If $P_{st} \neq C$, no path between core vertices can use C , and the restriction is a system on J . If $P_{st} = C$, contract the entire selected chain to the proxy edge st , obtaining a system on K .

For a core vertex v , consistency gives a cut k_v : paths from x_i to v enter the core through s for $i \leq k_v$ and through t for $i > k_v$. If $u \in P_{sv}$, then $k_v \leq k_u$; if $u \in P_{tv}$, then $k_u \leq k_v$. These implications define a preorder on the six core vertices.

Proposition 7.1 (Terminal-order core theorem). *For every neighborly consistent system on J or K , and every linear extension of its cut preorder after equivalent vertices are contracted, there is a positive rational inducing weight w such that*

$$\psi(v) = d_w(t, v) - d_w(s, v) \quad (7.2)$$

is strictly increasing along that linear extension.

Proof. This is the second finite computer-assisted proposition. There are 33 systems and 155 compatible block orders on J , and 144 systems and 321 compatible block orders on K . For each order, the rational system consists of all selected-path inequalities, edge lower bounds $w(e) \geq 1$, and a gap of at least one between consecutive ψ -blocks. The 476 static rational witnesses are stored in `terminal_order_core_certificates.json`. The independent verifier re-enumerates all 177 systems and all compatible block orders and checks every inequality exactly; it invokes no optimizer. $\square$

Lemma 7.2 (Long-ear extension). *Let w be any positive weighting that induces the restricted path system on the relevant core J or K . Suppose that $\psi(v) = d_w(t, v) - d_w(s, v)$ is weakly ordered by the actual cuts: whenever $k_u < k_v$, one has $\psi(u) < \psi(v)$, while vertices with the same cut may be placed in either order. Then positive weights can be assigned to a chain C of arbitrary length so as to induce all core-to-ear entrance choices, every direct or core-detour path between ear vertices, and all paths of the restricted core system.*

Proof. First assume $P_{st} \neq C$, so the core is J . Let $D = d_w(s, t)$, $p = k_t$, and $h = k_s$. The triangle inequality gives $-D \leq \psi(v) \leq D$, and cut compatibility gives $p \leq k_v \leq h$. Choose increasing right-side labels $z_{p+1} < \dots < z_m$ so that the values $\psi(v)$ with cut k lie in the corresponding open gap, with $z_{h+1} > D$ and $z_m > D$.

Let F be the set of ear pairs (i, j) , $i < j$, whose selected path detours through the core. Consistency gives

$$F \subseteq \{0, \dots, p\} \times \{h+1, \dots, m\}$$

and makes F expanding. Every left row is nonempty, and if $r_i = \min\{j : (i, j) \in F\}$, then $r_0 = h+1$ and the r_i are nondecreasing. Here the two assertions are forced rather than assumed: for $i \leq p = k_t$, the selected x_i - t path enters the core at s , so $(i, m) \in F$; and the definition $h = k_s$ says exactly that the first detour in row 0 occurs at $h+1$. Expanding closure then makes the row thresholds nondecreasing. Choose increasing $H_i \in (z_{r_i-1}, z_{r_i})$, with $H_0 = D$, and put

$$T = D + z_m, \quad z_i = H_i - T \quad (0 \leq i \leq p). \quad (7.3)$$

Then $z_0 = -z_m$ and

$$(i, j) \in F \iff z_j - z_i > T. \quad (7.4)$$

Assign

$$w(x_{i-1}x_i) = \frac{z_i - z_{i-1}}{2}. \quad (7.5)$$

The route from x_i to a core vertex enters through s exactly when $z_i \leq \psi(v)$, realizing k_v ; and (7.4) says that a core detour beats the direct ear segment exactly for the selected pairs. Since $z_m > D$, the ear creates no shorter route between core vertices.

If $P_{st} = C$, use the K -certificate and let $L = w(st)$. Here cuts $-1, m$ may occur. Consistency gives $k_v = -1 \Rightarrow \psi(v) = -L$ and $k_v = m \Rightarrow \psi(v) = L$. Delete the proxy edge, choose $-L = z_0 < \dots < z_m = L$ with all other ψ -blocks in their prescribed gaps, and use (7.5). The chain has total weight L , so it replaces the proxy edge without altering core distances: a path between old vertices that enters the new chain must enter at one terminal and leave at the other, and is therefore in weight-preserving correspondence with a path using the proxy edge. Likewise every selected core path using st expands to the whole chain. No ear pair can detour through the core, because such a path would prescribe an s - t subpath different from C . $\square$

Corollary 7.3. *Every $H_q = P_4(2, q, 2)$ is neighborly metrizable.*

Proof. Choose from Proposition 7.1 a core certificate whose block order extends the actual cut order, then apply Lemma 7.2. $\square$

Corollary 7.4. *The graph $H_3 = P_4(2, 3, 2)$ is metrizable.*

Proof. Neighborly systems are metrizable by Corollary 7.3. In a non-neighborly system, choose an edge $e = xy$ with $P_{xy} \neq e$ and delete it by Lemma 2.2. Every nontrivial block of $H_3 - e$ is a cycle or a theta subdivision. In kernel-edge order 01, 02, 03, 12, 13, 23, breaking the six corridors gives, respectively, theta-arm triples

$$(2, 2, 4), (1, 3, 5), (3, 3, 3), (1, 3, 3), (1, 3, 5), (2, 2, 4).$$

Indeed, after corridor ij is broken, the other two kernel vertices are the theta terminals and the three arms are their direct corridor and the two two-corridor routes through i and j . Thus each block has an arm of length at most two or is $\Theta_{3,3,3}$, and is metrizable by (1.1). Restore e with sufficiently large weight using Lemma 2.2. $\square$

Proof of Theorem 1.1. Every graph in (1.3) is non-metrizable, by [4], (1.1), and the certificate (3.2). Topological-minor closure gives necessity.

Conversely, Proposition 4.1 reduces an avoiding subdivision to the families of Theorems 6.3 and 6.7, the two finite graphs in Proposition 6.8, or $P_4(2, 3, 2)$. The last graph is metrizable by Corollary 7.4. These results give sufficiency.

It remains to justify the asserted minimality of Z_Δ . By Lemma 2.6, first consider a proper minor that retains all six kernel corridors. A first unit shortening of the length-three triangle edge gives the positive triangle core $(2, 2, 2)$; a first shortening of either length-two triangle edge leaves at most two long kernel edges. Both are positive by Proposition 6.8 and Theorem 6.3. Every further coordinate shortening is a topological minor of one of these first shortenings and is therefore metrizable.

If a kernel corridor is broken, the only nontrivial block is contained in the theta graph shown in the corresponding row of Table 2; pendant pieces are trees. The entries are the three arm lengths between the endpoints of the broken corridor. Every row is metrizable by (1.1). Breaking

Table 2: Theta blocks obtained by breaking a kernel corridor of Z_Δ .

corridor	arms	corridor	arms	corridor	arms
01	(2, 3, 4)	02	(2, 2, 5)	03	(2, 2, 5)
12	(1, 3, 4)	13	(1, 3, 4)	23	(1, 3, 3)

additional edges or deleting an internal vertex only takes topological minors of these theta blocks and adds tree pieces. Topological-minor closure and cut-vertex gluing therefore imply that every proper topological minor of Z_Δ is metrizable. $\square$

Proof of Theorem 1.2. The four Z -graphs and N_{233} have non-metrizable neighborly systems. By Lemma 2.4, every coordinate subdivision of one is a neighborly obstruction. Conversely, Proposition 4.2 leaves the at-most-two-long-edge families, the triangle and star finite cores, $P_4(p, 2, r)$, and H_q . They are handled respectively by Theorems 6.3 and 6.7, Proposition 6.8, and Corollary 7.3. This proves the classification. $\square$

Corollary 7.5 (Recognition from the six lengths). *Ordinary and neighborly metrizability of a K_4 -subdivision can be decided from its six kernel lengths using a constant number of integer comparisons, independent of the number of subdivision vertices.*

Proof. For each of the 24 kernel relabellings, compare the length vector with the four Z -vectors and, in the neighborly case, with the vector of N_{233} . By Lemma 2.5, this detects exactly the coordinate obstructions. For the ordinary theta obstruction, choose each unordered pair i, j of kernel vertices. If k, l are the other two vertices, the three internally disjoint i - j corridors have lengths

$$\ell_{ij}, \quad \ell_{ik} + \ell_{kj}, \quad \ell_{il} + \ell_{lj}.$$

After sorting, they contain $\Theta_{3,3,4}$ exactly when they dominate $(3, 3, 4)$. Six terminal pairs and finitely many vector comparisons prove the claim. $\square$

8 Exact computation and reproducibility

The finite propositions use the following complete encoding. We include its proof because it is the link between the mathematical finite statements and the enumerated index sets.

Lemma 8.1 (Rooted-tree encoding). *Let G be connected. Sending a neighborly consistent path system $\mathcal{P}$ to*

$$T_x = \bigcup_{y \neq x} P_{xy} \quad (x \in V(G))$$

is a bijection between neighborly consistent systems and tuples of spanning trees (T_x) satisfying

- (i) every edge incident with x belongs to T_x ; and*
- (ii) the unique x - y paths in T_x and T_y coincide for every pair x, y .*

Proof. For a consistent system, the union of the paths rooted at x is connected. It contains no cycle: on a first divergence and reconvergence along a cycle, two root paths would give different selected subpaths between the divergence and reconvergence vertices. Hence the union is a spanning tree. Neighborliness forces every incident edge into it, and symmetry of P_{xy} gives (ii).

Conversely, define P_{xy} as the common path in T_x, T_y . If $u, v \in P_{xy}$, condition (ii) for (x, u) puts the x - u subpath of P_{xy} into T_u , and condition (ii) for (y, u) puts the y - u subpath into T_u . Their union contains all of P_{xy} , so the unique u - v path in T_u is the corresponding subpath. The same argument puts that subpath in T_v ; applying (ii) to (u, v) proves that it is P_{uv} . Thus the system is consistent. Condition (i) makes every graph edge the selected path between its endpoints. The two constructions are inverse because a rooted tree is the union of its root-to-vertex paths. $\square$

The verifier implements the following depth-first join. Canonical path tuples, not solver output, are the keys, so the enumeration is both exhaustive and duplicate-free by Lemma 8.1.

```

for each root x:
  R[x] = spanning trees containing every edge incident with x
search(partial tuple):
  choose the next root x
  for T in R[x]:
    reject T if any already fixed x-y path disagrees
    otherwise recurse
  at a complete tuple, canonicalize all P[x,y] and emit once

```

For each complete system and each pair u, v , the verifier enumerates every simple u - v path Q and checks

$$w(P_{uv}) \leq w(Q), \quad w(e) \geq 1, \quad (8.1)$$

using Python's exact `Fraction` arithmetic. The discovery scripts use linear programming only to locate a candidate rational point. The delivered verification scripts do not call an optimizer: they reconstruct the entire finite index set and validate the stored fractions.

Table 3: Static exact certificate archives.

archive	stored systems	proof-critical instances	exact witnesses
ordinary core archive	4,354	1,052	4,354
terminal-order cores J, K	177	476	476

The ordinary archive deliberately retains the 3,302 systems on $P_4(2, 3, 2)$ as a regression set. They are not a premise of either main theorem: Lemma 7.2 and Corollaries 7.3 and 7.4 prove that case symbolically. Only the 510 triangle-core and 542 star-core records certify Proposition 6.8.

Each record is self-contained. Its schema is given in Table 4; paths and edges use integer vertex labels, and every weight is stored as a reduced numerator/denominator string. The **system** field, rather than an enumeration index, is the primary mathematical identifier, so a change in search order cannot silently align a certificate with the wrong system.

Table 4: Fields in the static JSON certificate records.

field	meaning
case/core	named finite graph and, where applicable, its kernel length signature
edges	canonical ordered edge list; this fixes the order of weights
system	complete map $u-v \mapsto$ selected vertex path
block_order	ordered terminal-potential blocks for a terminal-core instance; absent from ordinary records
weights	positive reduced rational weight for every edge

The optimization-free verification has four logically separate parts:

- (1) `verify_ztriangle.py` and `verify_neighborly_n233.py` check the two new integer obstruction identities;
- (2) `verify_ordinary_finite_core_certificates.py` re-enumerates all 4,354 archived systems and verifies every simple-path inequality; the first 1,052 records are proof-critical and the remaining 3,302 form the independent H_3 regression set;
- (3) `verify_terminal_order_core_certificates.py` re-enumerates all 476 system/order instances and checks both shortest paths and the terminal-potential gaps;
- (4) `verify_fixed_margin_extension.py` checks the explicit insertion formula on all 2,035 expanding threshold sequences of the documented bounded regression range.

The identities in item (1) are fully displayed in (3.2) and (3.4), so their verification is hand-checkable and the scripts are optional audits. The proof-critical part of item (2) and all of item (3) certify the two finite propositions. Item (4), the last 3,302 records in item (2), and the long-ear runs below are regression tests for symbolic lemmas and are labelled as such. Running `bash run_verification.sh` executes all optimization-free certificate checks; the SHA-256 manifest detects any change to a source, archive, or compiled manuscript.

Two further structure-only audits test the case distinctions without supplying any theorem premise. The adjacent-chain script checked 1,952 neighborly systems on the documented small instances, and the split-bridge script checked all 8,135 systems for $(p, r) = (2, 2), (3, 2), (2, 3)$. In particular, the latter found 418 systems that actually select the fifth route W , while reporting no violation of the five-route list, mode separation, entrance-prefix rules, rectangular normal forms, or outward-safety color sets. These counts explain why the fifth route is retained explicitly in the symbolic proof.

As a regression test independent of the infinite proof, the long-ear constructor was applied to every system on H_3, H_4, H_5, H_6 , respectively 3,302, 9,695, 24,942, and 59,053 systems. It selected an archived core witness, constructed all ear weights by (7.3)–(7.5), recomputed exact all-pairs distances, and passed all 96,992 systems. These tests audit endpoint conventions and formulas; they are not used in place of Lemma 7.2.

Online Resource 1 contains the source, both JSON certificate files, two completeness verifiers that call no optimizer, the integer verifiers for Z_Δ and N_{233} , the long-ear constructor, and a SHA-256 manifest. A short README gives one-command reproduction instructions and distinguishes certificate verification from optional LP-based archive regeneration.

9 Audit of chordless competitors

For clarity, Table 5 records where every possible chordless competitor is handled. This is not an additional assumption: it is a compact index to the induced-path inspections in the proofs. Once these rows are checked, Lemma 2.3 disposes of every path with a chord.

Table 5: Exhaustion ledger for positive constructions.

family	chordless competitors	point of realization
opposite suspended chains	four one-bridge cross routes; a direct chain segment or one of two endpoint detours	four-corner representation and the fixed-margin outward-safe extension in Theorem 6.3
adjacent suspended chains	S, T, U, V ; two entrances from x ; direct or one of the two exterior same-chain routes	route exhaustion and common prefix thresholds in Lemma 6.1; midpoint positivity and detours in Lemma 6.2
split bridge $P_4(p, 2, r)$	four one-bridge routes plus W ; the six s -root routes in (6.17); two same-chain exterior routes	exhaustive list and exact interval condition (6.21) in Lemma 6.4; global $Q/W/N$ normal forms in Lemma 6.5; simultaneous scalar calibration in Lemma 6.6
long ear H_q	entrance through s or t ; direct ear segment or one core detour; old-core paths with or without the proxy edge	terminal potential cuts and difference threshold in Lemma 7.2
finite bases	every simple path between every unordered vertex pair on the triangle and star cores; every compatible terminal-potential order on J, K	exact all-simple-path comparison in the verifiers for Propositions 6.8 and 7.1

The split-bridge line deserves one final check. An induced cross path cannot alternate between the chains twice: after the first two bridge crossings, one of a_0b_0, a_0b_r, a_pb_r is a chord unless the path is exactly the five-cycle complement W . An induced path rooted at s must first use sa_p or sb_0 ; after that choice, the same chord test leaves exactly the three routes in the relevant half of (6.17). Thus the ledger includes the previously easy-to-miss fifth cross route and both diagonal root routes.

The symbolic part of the proof is constructive in the following concrete sense. From the selected path table one reads the finite cut positions and Ferrers relations; the order-insertion proofs then return rational labels, because every required open interval has rational endpoints after a common rational scaling. The displayed formulas convert these labels to positive rational edge weights. In the two bounded ordinary cores and the two terminal cores, the finite base operation is a lookup by the canonical **system** key in the static archives. The bounded graph $P_4(2, 3, 2)$ uses the symbolic long-ear construction rather than such a lookup.

10 Concluding remarks

The two classifications show that ordinary and neighborly metrizable split already within a fixed four-branch-vertex topological type. Ordinary metrizable is governed by five topological

obstructions; neighborly metrizable discards the theta obstruction but gains the coordinate obstruction $P_4(2, 3, 3)$. The Ferrers and long-ear constructions provide humanly checkable certificates of positive metrizable for every infinite family, complementing the non-metrizable certificates of [4] and the structural results of [2].

Two broader questions remain. First, the full obstruction sets $\mathcal{F}_M$ and $\mathcal{F}_{SM}$ are still unknown outside the K_4 topological type. Second, the neighborly property is not topological-minor closed, so its general structure will require a containment notion that records which chords force one-edge selected paths.

Statements and Declarations

Funding. This work was supported by the Natural Science Foundation of Shandong Province under Grant ZR2024MA073. The funder had no role in the design of the study; in the collection, analysis, or interpretation of the certificate data; in the writing of the manuscript; or in the decision to submit it for publication.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Author contributions. Guangfu Wang: conceptualization, methodology, formal analysis, software, validation, investigation, visualization, writing—original draft, and writing—review and editing. The author read and approved the final manuscript.

Data, materials, and code availability. Online Resource 1 and the public reproducibility repository accompanying this article contain the complete static certificate archives, the optimization-free exact verifiers, the optional archive-regeneration programs, the structural regression scripts, the manuscript source, a machine-readable citation file, archival metadata, and SHA-256 manifests. The command `bash run_verification.sh` re-enumerates all finite instances used in the two computer-assisted propositions and checks every stored witness using only integer or exact rational arithmetic. It retains the 3,302 H_3 records solely as a regression test for the independent symbolic long-ear proof. No proprietary data or software are required. The versioned archive supplied with the submission is identical to the tagged repository release; its permanent DOI will be inserted at proof stage if the archival deposit is not available before submission.

Ethics approval and consent to participate or publish. Not applicable. The study uses no human participants, personal data, animals, or third-party copyrighted experimental material.

Use of generative AI. During preparation of this manuscript, the author used ChatGPT for proof-structure brainstorming, adversarial review, and language editing. The tool was not treated as an author and did not replace mathematical proof or exact verification. The author reviewed and edited all generated material and accepts full responsibility for the content of the publication.

Acknowledgments. The author thanks the developers of the open-source software used for exact verification.

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